



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 7

TITLE: Implement a Java program to demonstrate inheritance and interfaces using suitable

real-world examples.

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

1.Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

Main.java



Run

```
1 import java.util.Scanner;
2
3 class Shape {
4     void area() {
5     }
6 }
7
8 class Circle extends Shape {
9     double radius;
10
11     Circle(double radius) {
12         this.radius = radius;
13     }
14
15     void area() {
16         System.out.println("Area of Circle: " + (3.14 * radius * radius));
17     }
18 }
19
20 class Rectangle extends Shape {
21     double length, width;
22
23     Rectangle(double length, double width) {
24         this.length = length;
25         this.width = width;
26     }
27
28     void area() {
29         System.out.println("Area of Rectangle: " + (length * width));
30     }
31 }
```

```
33 ▾ class Main {  
34 ▾     public static void main(String[] args) {  
35         Scanner sc = new Scanner(System.in);  
36  
37         System.out.print("Enter Radius: ");  
38         double r = sc.nextDouble();  
39  
40         System.out.print("Enter Length: ");  
41         double l = sc.nextDouble();  
42  
43         System.out.print("Enter Width: ");  
44         double w = sc.nextDouble();  
45  
46         Circle c = new Circle(r);  
47         Rectangle rect = new Rectangle(l, w);  
48  
49         c.area();  
50         rect.area();  
51  
52         sc.close();  
53     }  
54 }
```

OUTPUT

Output

Enter Radius: 5
Enter Length: 6
Enter Width: 8
Area of Circle: 78.5
Area of Rectangle: 48.0

=== Code Execution Successful ===

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

Main.java

 Share 

```
1 interface Product {
2     void display();
3 }
4
5 class ProductDetails {
6     int id;
7     String name;
8     double price;
9
10    ProductDetails(int id, String name, double price) {
11        this.id = id;
12        this.name = name;
13        this.price = price;
14    }
15 }
16
17 class Electronic extends ProductDetails implements Product {
18    Electronic(int id, String name, double price) {
19        super(id, name, price);
20    }
21
22    public void display() {
23        System.out.println("Electronic Product");
24        System.out.println("ID: " + id);
25        System.out.println("Name: " + name);
26        System.out.println("Price: " + price);
27    }
28 }
29
30 class Clothing extends ProductDetails implements Product {
31    Clothing(int id, String name, double price) {
```



```
32         super(id, name, price);
33     }
34
35     public void display() {
36         System.out.println("Clothing Product");
37         System.out.println("ID: " + id);
38         System.out.println("Name: " + name);
39         System.out.println("Price: " + price);
40     }
41 }
42
43 class Grocery extends ProductDetails implements Product {
44     Grocery(int id, String name, double price) {
45         super(id, name, price);
46     }
47
48     public void display() {
49         System.out.println("Grocery Product");
50         System.out.println("ID: " + id);
51         System.out.println("Name: " + name);
52         System.out.println("Price: " + price);
53     }
54 }
55
56 class Main {
57     public static void main(String[] args) {
58         Electronic e = new Electronic(101, "Laptop", 95000);
59         Clothing c = new Clothing(102, "T-Shirt", 1260);
60         Grocery g = new Grocery(103, "Rice", 450);
61
62         e.display();
63         System.out.println();
64
65         c.display();
66         System.out.println();
67
68         g.display();
69     }
70 }
```

OUTPUT

```
Output

Electronic Product
ID: 101
Name: Laptop
Price: 95000.0

Clothing Product
ID: 102
Name: T-Shirt
Price: 1260.0

Grocery Product
ID: 103
Name: Rice
Price: 450.0

=== Code Execution Successful ===
```

Conclusion:

This experiment demonstrated the use of **inheritance** and **interfaces** in Java to achieve code reusability, abstraction, and polymorphism. Inheritance enabled derived classes to reuse the properties and methods of a base class, reducing code duplication, while interfaces provided a common contract that implementing classes followed. Together, these object-oriented concepts improved the flexibility, maintainability, and modularity of the program, making it easier to design efficient and scalable real-world applications.